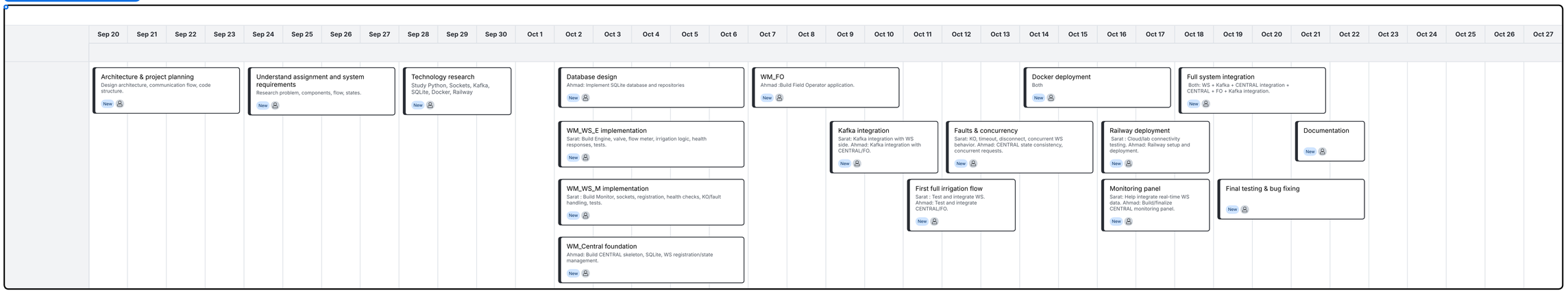
Timeline


Sep 20
Sep 21
Sep 22
Sep 23
Sep 24
Sep 25
Sep 26
Sep 27
Sep 28
Sep 29
Sep 30
Oct 1
Oct 2
Oct 3
Oct 4
Oct 5
Oct 6
Oct 7
Oct 8
Oct 9
Oct 10
Oct 11
Oct 12
Oct 13
Oct 14
Oct 15
Oct 16
Oct 17
Oct 18
Oct 19
Oct 20
Oct 21
Oct 22
Oct 23
Oct 24
Oct 25
Oct 26
Oct 27
Architecture & project planning
Design architecture, communication flow, code structure.
New
Understand assignment and system requirements
Research problem, components, flow, states.
New
Technology research
Study Python, Sockets, Kafka, SQLite, Docker, Railway
New
Database design
Ahmad: Implement SQLite database and repositories
New
WM_WS_E implementation
Sarat: Build Engine, valve, flow meter, irrigation logic, health responses, tests.
New
WM_WS_M implementation
Sarat : Build Monitor, sockets, registration, health checks, KO/fault handling, tests.
New
WM_Central foundation
Ahmad: Build CENTRAL skeleton, SQLite, WS registration/state management.
New
WM_FO
Ahmad :Build Field Operator application.
New
Kafka integration
Sarat: Kafka integration with WS side. Ahmad: Kafka integration with CENTRAL/FO.
New
First full irrigation flow
Sarat : Test and integrate WS. Ahmad: Test and integrate CENTRAL/FO.
New
Faults & concurrency
Sarat: KO, timeout, disconnect, concurrent WS behavior. Ahmad: CENTRAL state consistency, concurrent requests.
New
Docker deployment
Both
New
Railway deployment
Sarat : Cloud/lab connectivity testing. Ahmad: Railway setup and deployment.
New
Monitoring panel
Sarat: Help integrate real-time WS data. Ahmad: Build/finalize CENTRAL monitoring panel.
New
Full system integration
Both: WS + Kafka + CENTRAL integration + CENTRAL + FO + Kafka integration.
New
Final testing & bug fixing
New
Documentation
New